

# Determining the Favorability of Intraspecific Cooperation over Interspecific Cooperation by Darwinian Evolution Using a Genetic Algorithm

## Introduction

Past research has observed, even though evolution theory suggests that the motivating factor for evolution is competition, intraspecific cooperative behavior paradoxically evolves abundantly (Li et al., 2025). This situation becomes even more unclear in the case of mutualism which sees the evolution interspecific cooperation. Game theory can emulate the underlying risk assessment of making cooperative choices (L.F.P. Salles et al., 2024).

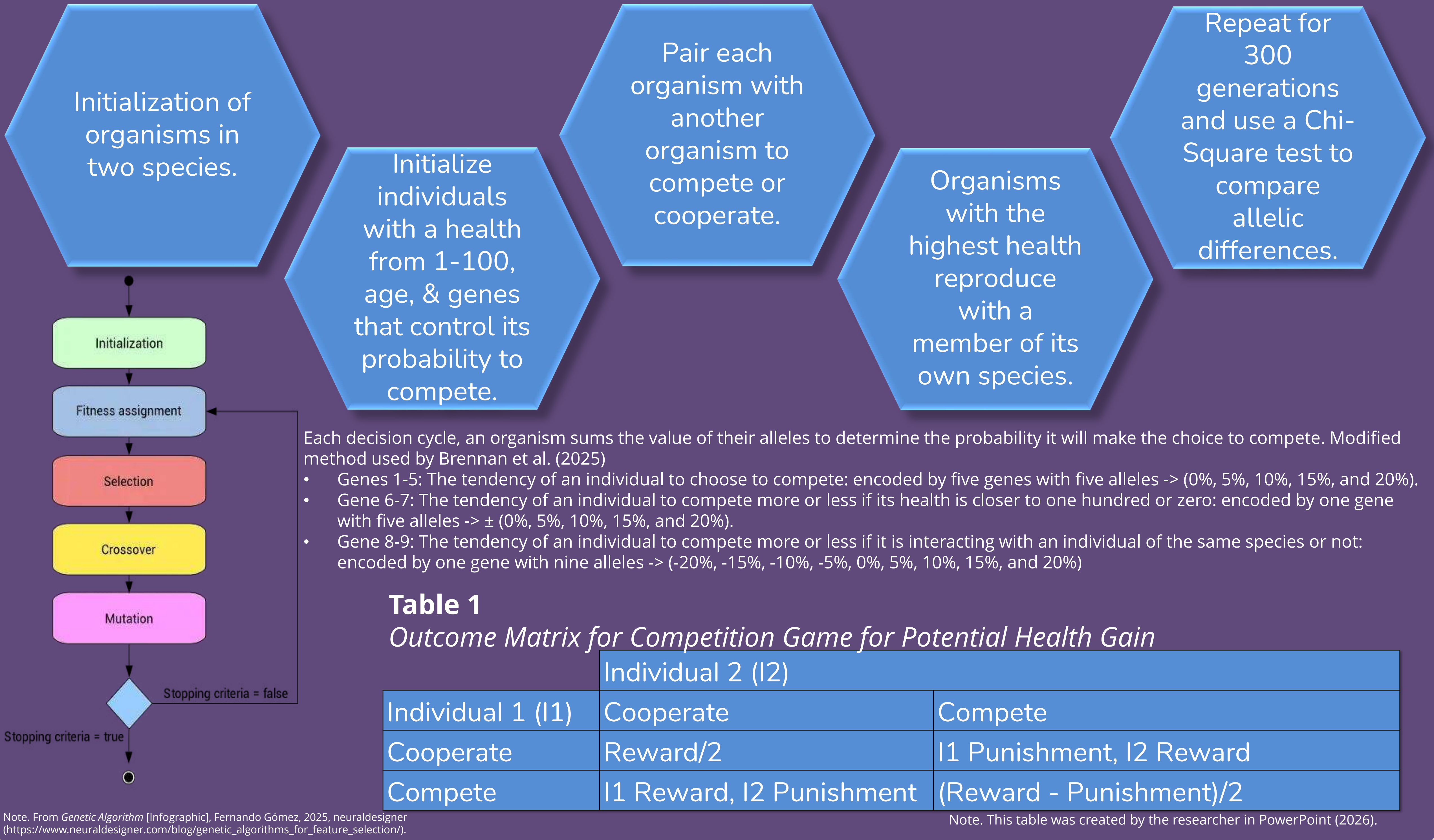


Intraspecific Competition  
Note. From Community Ecology [Photograph], Wizeprep, n.d., Wizeprep (Retrieved February 20, 2026, from <https://www.wizeprep.com/online-courses/5762/chapter/34/core/4/1>). CC BY-SA 4.0

## Purpose

The purpose of this research is to determine if interspecies cooperation is favored by Darwinian evolution and certain environmental constraints. This can provide insight on how isolated factors, such as reproductive isolation, influences cooperation. This is especially important considering that Darwinian evolution would not favor individuals making sacrifices for another as that would reduce chances of reproduction for the individual making a sacrifice.

## Methodology



## Results

Greater punishment than reward:

Table 2

Allelic differences of Competitive Genes

| Allele | Intraspecific | Interspecific |
|--------|---------------|---------------|
| -20%   | 40            | 33            |
| -15%   | 45            | 48            |
| -10%   | 59            | 70            |
| -5%    | 98            | 63            |
| 0%     | 63            | 153           |
| 5%     | 146           | 297           |
| 10%    | 372           | 331           |
| 15%    | 1039          | 527           |
| 20%    | 638           | 978           |

Chi-Square value: **339.61**

Degrees of Freedom: 16

Critical value: **26.296** (with significance of 0.05)

Chi-Square value > Critical value:  
**Reject null hypothesis;** there is evidence of difference between the interspecies competition alleles and intraspecies competition alleles.

Note. This table was created by the researcher in PowerPoint (2026).

Table 3

Tallied Choices to Compete or Cooperate

| Intraspecific Competition | Intraspecific Cooperation | Interspecific Competition | Interspecific Cooperation |
|---------------------------|---------------------------|---------------------------|---------------------------|
| 108967                    | 9993                      | 110488                    | 9794                      |

Ratio of cooperation over total number of intraspecies interactions: **0.084**  
Ratio of cooperation over total number of interspecies interactions: **0.081**

Note. This table was created by the researcher in PowerPoint (2026).

## Results

Equal punishment and reward:

Table 4

Allelic differences of Competitive Genes

| Allele | Intraspecific | Interspecific |
|--------|---------------|---------------|
| -20%   | 40            | 34            |
| -15%   | 18            | 33            |
| -10%   | 66            | 76            |
| -5%    | 60            | 62            |
| 0%     | 96            | 121           |
| 5%     | 253           | 268           |
| 10%    | 461           | 340           |
| 15%    | 547           | 615           |
| 20%    | 959           | 951           |

Chi-Square value: **31.24**

Degrees of Freedom: 16

Critical value: **26.296** (with significance of 0.05)

Chi-Square value > Critical value:  
**Reject null hypothesis;** there is evidence of difference between the interspecies competition alleles and intraspecies competition alleles.

Note. This table was created by the researcher in PowerPoint (2026).

Table 5

Tallied Choices to Compete or Cooperate

| Intraspecific Competition | Intraspecific Cooperation | Interspecific Competition | Interspecific Cooperation |
|---------------------------|---------------------------|---------------------------|---------------------------|
| 113378                    | 5106                      | 114004                    | 5026                      |

Ratio of cooperation over total number of intraspecies interactions: **0.043**  
Ratio of cooperation over total number of interspecies interactions: **0.042**

Note. This table was created by the researcher in PowerPoint (2026).

## Conclusion

- Considering Table 3 & 5, overall, there was a large amount of competition. This may be explained by the research done in 2025 by Li et al. and Brennan et al. which saw populations with many competitors saw an encouragement of further competitive behavior.
- Unlike that previous research, this project compares the difference between intra/interspecific competition.
- With greater punishment than reward, there was a large statistical difference in the allelic differences between the intraspecific competitive inclination gene and the interspecific one (Table 2).
- With equal punishment and reward, there was a smaller but still significant difference (Table 4).
- There was a slight positive difference in the ratio of cooperation over total number of intraspecies and intraspecies interactions (Table 3 & 5) indicating a slight favorability towards intraspecies cooperation amplified by the risk of competing.

## Future Work

Future work can introduce other factors that may cause speciation and consequently affect interspecies competition into the simulation. Furthermore, the application of this simulation to empirical studies can provide insight into its significance.

## References

- Brennan, C., Chassy, P., & Cole, J. (2025). Evolution of confrontation and cooperation in simple organisms as a function of environmental resources and cost of a conflict. *Expert Systems*, 42(1), 1–14. <https://doi.org/10.1111/exsy.13527>.
- Fernando Gómez. (2025). *Genetic Algorithm* [Infographic]. neuraldesigner. Retrieved February 20, 2026, from [https://www.neuraldesigner.com/blog/genetic\\_algorithms\\_for\\_feature\\_selection/](https://www.neuraldesigner.com/blog/genetic_algorithms_for_feature_selection/) <https://doi.org/10.1016/j.amc.2025.129497>.
- Li, L., Lv, J., Ma, L., & Ruan, J. (2025). Community reciprocity promotes cooperation in a network evolutionary prisoner's dilemma game. *Applied Mathematics and Computation*, 502(129497).
- Salles, L. F. P., Aloizio Martinez de Aguiar, M., & Marquitti, F. M. D. (2024). Evolution of cooperation in a two-species system with a common resource pool. *Journal of Theoretical Biology*, 577(111670). <https://doi.org/10.1016/j.jtbi.2023.111670>.
- Wizeprep. (n.d.). *Community Ecology* [Photograph]. Wizeprep. Retrieved February 20, 2026, from <https://www.wizeprep.com/online-courses/5762/chapter/34/core/4/1>